DEPI | DATA ANALYSIS FINAL PROJECT

Requirements Gathering

Stakeholder Needs, System Requirements, and Validation Framework

Project: Interactive Power BI Urban Intelligence Dashboard
Focus: Egypt's New Administrative Capital

1. Project Overview and Requirements Gathering Approach

Project background. The project is a student-developed Power BI dashboard that organizes available urban indicators for Egypt's New Administrative Capital. Users will be able to explore city conditions, compare compatible measures, identify potential areas of concern, and interpret evidence to support informed discussion. The dashboard is an academic decision-support prototype; it is not commissioned, adopted, or approved by a government authority and does not make official decisions.

Business problem and purpose. Urban information may be distributed across sources, use different definitions, and cover different periods or geographic boundaries. Requirements gathering establishes what users need, what data can credibly support, and how the dashboard will demonstrate those needs before development. It reduces scope creep and aligns the work with project objectives.

Primary objectives. (1) Consolidate traceable urban indicators. (2) Present a clear overview across five domains. (3) enable supported filtering and comparisons. (4) communicate definitions, sources, dates, and limitations. (5) validate data, calculations, interactions, and usability.

Scope

In scope: collect and prepare relevant datasets; define indicators and record sources; build interactive Power BI pages; present supported comparisons and descriptive insights; test accuracy, calculations, filters, navigation, and usability; document limitations and evidence-informed interpretations.

Out of scope unless approved and feasible: production government deployment; direct live-system integration; confidential/restricted personal data; automatic policy decisions; real-time monitoring without suitable feeds; predictive modelling or scenario simulation without appropriate data, methods, and validation.

Requirements-gathering approach

The process follows the supplied Requirements Gathering Process Guide: identify stakeholders; plan the approach; conduct elicitation; document findings; analyse and prioritise; verify and validate; and manage changes. Proposed techniques are source/document review, short interviews or consultations, optional surveys, and prototype walkthroughs. Activities are plans, not claims of completed consultations.

Assumptions and constraints. Public or otherwise authorised data may be available, but no domain is guaranteed coverage. Time, student resources, Power BI capabilities, inconsistent definitions, and lack of confirmed end users constrain scope. Only indicators with sufficiently clear sources, definitions, geography, and reporting periods should be implemented.

Project Objective	Business Need	Expected Outcome	Verification Method
Consolidate urban evidence	Information is distributed across sources	Traceable, prepared indicator dataset	Source and field audit
Enable exploration	Users need a coherent city overview	Overview and domain pages	Page/content checklist
Support comparison	Comparable measures aid interpretation	Filtered comparisons where valid	Comparison test cases
Promote responsible use	Context and limitations affect meaning	Definitions, sources, dates and caveats	Metadata and caveat review

2. Proposed Stakeholder Analysis

This is a planning analysis of anticipated stakeholder groups. Potential external users are hypothetical; no interview, commitment, approval, or participation is implied. Influence and interest ratings are initial estimates and should be revisited if actual reviewers become available.

Stakeholder	Role in Project	Main Needs and Expectations	Influence	Interest	Engagement Approach
Project team	Collects data; analyses, builds, documents and tests	Feasible scope, clear definitions, reliable data, manageable workload	High	High	Frequent working reviews; maintain requirements and issue logs
DEPI instructor / evaluator	Academic reviewer of method, implementation and presentation	Clear objectives, sound analysis, evidence, traceability and honest limitations	High	High	Planned milestone walkthroughs; request feedback and record decisions
Potential urban decision-makers / planners	Proposed end users	Summarised indicators, comparisons and evidence for prioritisation	Medium*	High	Validate key questions and interpretation through consultation if accessible
Potential public-service / transport planners	Proposed domain users	Relevant service, mobility and infrastructure measures	Medium*	High	Review domain indicators and geographic comparability if accessible
Potential environment / sustainability stakeholders	Proposed domain users	Credible environmental indicators and trends	Medium*	Medium–High	Review definitions, units, periods and caveats
Potential public / general users	Proposed audience	Understandable overview and plain-language labels	Low–Medium*	Medium	Optional usability survey or prototype walkthrough
Data providers / source organisations	Publish or maintain source datasets	Accurate attribution, compliant use and contextual interpretation	Medium–High*	Medium	Review metadata, licences, methodology and update dates

*Influence is provisional and reflects potential control over data, review, or adoption—not confirmed authority over this student project.

Power–Interest Matrix

Engagement Group	Proposed Stakeholders	Communication Approach
Manage closely (high power / high interest)	Project team; DEPI instructor/evaluator	Regular progress reviews, requirements walkthroughs, issue resolution and validation checkpoints
Keep satisfied (high power / lower interest)	Relevant data-source organisations, where their permissions or terms affect use	Respect source conditions; request clarification only when needed; share concise questions
Keep informed (lower power / high interest)	Potential planners and domain users; interested public users	Share prototype snapshots and concise explanations; invite optional feedback
Monitor (lower power / lower interest)	Other indirectly affected audiences	Provide final summary and accessible context; engage if interest increases

The matrix guides the level and frequency of communication. It does not establish stakeholder authority or approval. Validation will use available reviewers; if a proposed user group cannot be reached, the limitation will be recorded and instructor review and testable acceptance criteria will be used instead.

3. Requirements Elicitation Plan

Elicitation will focus first on underlying information needs and decisions, rather than selecting visualisations prematurely. Findings, requests, constraints, and unresolved questions will be recorded in a central requirements register.

3.1 Document and data-source review

Review official or otherwise credible datasets, publications, indicator definitions, geographic coverage, reporting periods, units, update frequency, licensing, and methodology. Record availability and quality limitations before committing an indicator to the dashboard.

3.2 Interviews and consultations

If suitable reviewers, potential users, or subject-matter experts are available, conduct short structured conversations. Example questions: What urban question are you trying to answer? Which indicators would help? What geographic or time comparisons are meaningful? What context or caveats must be visible? What would make a chart misleading or difficult to use? These are proposed questions; no interviews are claimed as completed.

3.3 Surveys and feedback

An optional, proportionate survey may gather preferences for page navigation, labels, chart formats, filters, and usability. Use only if suitable participants are available; report the number and nature of responses if conducted, and do not generalise beyond the sample.

3.4 Prototyping and walkthroughs

Prepare a low-fidelity page wireframe or early Power BI prototype. Ask reviewers to complete realistic tasks, identify unclear labels or missing functionality, and explain interpretation difficulties. Record each comment, its source, decision, affected requirement, and resulting change.

Technique	Purpose	Target Participants or Sources	Information to Collect	Expected Output	Timing
Source review	Test feasibility and credibility	Public/authorised datasets, publications, metadata	Definitions, units, geography, period, licence, gaps	Source inventory and indicator feasibility log	Before indicator baseline; repeat during preparation
Interviews / consultations	Understand user questions	Available instructor, reviewers, potential users, experts	Decisions, indicators, comparisons, caveats	Interview notes and candidate requirements	Planning / early prototype
Optional survey	Collect broad usability preferences	Suitable available participants	Navigation, labels, charts, filters	Response summary and usability needs	After initial requirements or prototype
Prototype walkthrough	Check practical interpretation	Instructor/reviewers; potential users if available	Task completion, confusion, missing features	Feedback log and revised requirements	After core pages are drafted

4. User Stories and Use Cases

The following stories describe proposed user needs. Priorities are initial MoSCoW-aligned estimates and depend on data feasibility.

ID	User Role	User Story	Priority	Acceptance Criteria
US-01	General user	As a general user, I want to explore a city overview, so that I can understand the available urban indicators.	Must	Overview displays available core indicators with labels, units and reporting context; unavailable measures are not shown as zero.
US-02	Mobility planner*	As a mobility planner, I want to compare supported transport indicators, so that I can explore differences across available areas or periods.	Should	Comparison appears only where definitions, units, geography and periods are compatible; caveats are visible.
US-03	Environment stakeholder*	As an environment stakeholder, I want to review supported sustainability measures, so that I can examine the available environmental evidence.	Should	Environmental page shows source, unit and period for each displayed measure; gaps are disclosed.
US-04	Public-service planner*	As a service planner, I want to explore available public-service indicators, so that I can review the distribution of selected services.	Should	Page presents only sourced service measures and states geographic coverage; filters update relevant visuals.
US-05	Economic / admin analyst*	As an analyst, I want to compare reliable economic or administrative indicators, so that I can examine patterns in the available data.	Could	Only comparable indicators are compared; no unsupported causal claim is displayed.
US-06	Dashboard user	As a dashboard user, I want to filter by supported category, location or period, so that I can focus on relevant records.	Should	Each available filter changes all intended visuals consistently and can be cleared/reset.
US-07	All users	As a user, I want to inspect an indicator's definition, source, unit and date, so that I can interpret it correctly.	Must	Metadata is visible on the page, tooltip, or source note for each key indicator where feasible.
US-08	All users	As a user, I want to see missing-data and comparability limitations, so that I do not misinterpret incomplete evidence.	Must	Missing values are distinguishable from zero; relevant limitations are displayed beside affected analysis.

*Proposed user role, not a confirmed participant.

Use Case UC-01 — Explore and Compare Urban Indicators

Primary actor: Dashboard user (e.g., proposed planner). Goal: explore an indicator and compare compatible categories, locations, or periods. Preconditions: report is open; relevant dataset and metadata are loaded; comparison dimensions are supported.

Main flow: (1) Open City Overview. (2) Select a domain page. (3) Choose an indicator and available filters. (4) Review chart/table and displayed unit, source and period. (5) Compare only supported categories or periods. (6) Clear filters or return to overview.

Alternative / exception: If data is unavailable, show an explicit unavailable/not-covered message; if definitions or periods are incompatible, suppress the comparison or show a caveat. Postconditions: user sees the filtered descriptive result and its context; no official decision is executed.

Use Case UC-02 — Inspect Source and Interpret an Indicator

Primary actor: Any dashboard user. Goal: understand what a displayed measure represents and its limitations. Preconditions: an indicator is displayed and source metadata is recorded.

Main flow: (1) Select or hover over the indicator. (2) Read definition, unit, source and reporting date/period. (3) Review coverage and known limitations. (4) Decide whether the measure is suitable for the intended comparison.

Alternative / exception: If metadata is incomplete, label the gap and avoid presenting the measure as fully validated. Postconditions: user has the available context needed for cautious interpretation.

5. Functional Requirements

FRs describe capabilities of the supporting data workflow and Power BI prototype. Priorities are proposed and may change after source feasibility review.

ID	Requirement Description	Business Rationale	Priority	Acceptance Criteria	Verification
FR-01	Import datasets from identified public or authorised sources.	Establish an evidence base.	Must	Each included dataset has a recorded source and import method; unsupported sources are excluded.	Source/import audit
FR-02	Clean and standardise relevant fields, types, categories and geographic labels.	Improve consistency.	Must	Documented preparation steps address identified duplicates, invalid types and inconsistent labels.	Power Query step review; sample checks
FR-03	Handle missing and invalid values transparently.	Avoid misleading results.	Must	Missing values remain distinguishable from numeric zero; treatment is documented per field.	Null/zero test cases
FR-04	Retain source, definition, unit and reporting-period metadata for indicators.	Enable traceability.	Must	Each key indicator has source and period; unit/definition are recorded where available.	Metadata checklist
FR-05	Define indicators and calculations consistently.	Support reproducible interpretation.	Must	Each calculated measure has a documented formula, aggregation and applicable grain.	Recalculate sample records
FR-06	Present a general City Overview page.	Provide an entry point.	Must	Overview displays selected supported indicators and links/navigation to available domain pages.	Page inspection
FR-07	Organise supported content into five proposed domains: overview, mobility, environment, public services, economy/admin.	Make exploration coherent.	Must*	Each adequately supported domain has a page/section; unsupported domains are labelled and scope-adjusted transparently.	Domain coverage checklist
FR-08	Provide suitable charts, tables and comparisons.	Make patterns interpretable.	Must	Each visual matches the data type; comparison is enabled only for compatible measures.	Visual and comparison review
FR-09	Allow supported filters and visual interactions.	Enable focused exploration.	Should	Test selections update intended visuals consistently; reset/clear works.	Filter interaction tests
FR-10	Provide clear navigation between report pages.	Support task completion.	Must	Every intended page is reachable and navigation controls work.	Navigation test
FR-11	Display indicator name, unit, source and date where feasible.	Provide context at point of use.	Must	Key visuals or linked notes expose available metadata; absent metadata is not invented.	Metadata display review
FR-12	Highlight descriptive patterns without implying unsupported causation.	Prevent overclaiming.	Must	Narrative labels describe observed values/patterns only; causal language requires separate evidence.	Content review
FR-13	Validate calculations, aggregations, relationships, filters and navigation; correct defects before submission.	Protect quality.	Must	Documented test cases pass or unresolved defects are disclosed and prioritised.	QA checklist and defect log

*FR-07 is conditional on data availability: a domain with inadequate evidence must be marked as limited/not covered rather than populated with unsupported indicators.

6. Non-Functional Requirements

ID	Quality Attribute	Requirement	Measurable Acceptance Criteria	Verification
NFR-01	Usability	Use clear navigation, understandable labels, consistent formatting and appropriate visuals.	A reviewer can locate the overview and each available domain page; key chart titles, axes and legends are understandable in walkthrough.	Walkthrough checklist
NFR-02	Usability / accessibility	Use readable text, adequate contrast and more than colour alone to convey meaning.	Text and labels remain legible in the intended presentation view; status distinctions also use labels/icons/patterns.	Visual inspection
NFR-03	Performance	Keep report loading and interactions reasonable on the development environment.	Provisional target: common page interactions respond within 5 seconds on the test device with the prepared dataset; record environment and observed result.	Timed manual test
NFR-04	Performance / maintainability	Use manageable data models and measures where feasible.	No avoidable duplicate tables/relationships or unnecessarily repeated calculations remain after model review.	Model and measure review
NFR-05	Reliability / accuracy	Ensure calculations and aggregations are correct.	For each key measure, independently recalculate at least one representative sample and reconcile totals within expected rounding.	Calculation test
NFR-06	Reliability / accuracy	Apply consistent missing/invalid-value treatment and reproducible preparation.	Documented transformation steps can be refreshed/repeated; nulls are not silently converted to zero.	Refresh and null-handling test
NFR-07	Traceability	Trace displayed indicators to source datasets and definitions.	Every key indicator has a source reference and reporting period recorded where available.	Source-to-visual audit
NFR-08	Security / privacy	Use public or authorised data; avoid unnecessary personal or sensitive information.	Dataset inventory records access basis/licence where available; no restricted personal data is intentionally included.	Data and permission review
NFR-09	Maintainability	Use clear names and document transformations, calculations and update steps.	Tables, fields, measures and pages use meaningful names; core transformations and measures have concise documentation.	Documentation review
NFR-10	Compatibility	Support the intended presentation environment.	Report opens and core interactions work in the agreed Power BI Desktop or presentation environment.	Environment smoke test

Numerical performance values are provisional targets, not achieved results; confirm them during testing and record device, dataset size, and Power BI environment.

7. Project Constraints, Assumptions, and Dependencies

Type	Item	Potential Effect on Project
Constraint	Limited student time and resources	Prioritise minimum viable pages and core indicators; defer optional enhancements.
Constraint	Data availability/accessibility	Some domains or indicators may be omitted or represented as data gaps.
Constraint	Inconsistent periods, boundaries or definitions	Restrict comparisons; document caveats and avoid invalid aggregation.
Constraint	Power BI and development-environment capabilities	Design visuals and refresh approach around available licences, features and hardware.
Constraint	No confirmed government-system access or end users	No live integration or claim of operational adoption; validate with available reviewers.
Constraint	Limited authoritative reference data	State validation limits and avoid claiming conclusions are officially verified.
Assumption	Suitable public/authorised sources can be identified	If false, revise indicator scope and record unavailable domains.
Assumption	Available Power BI environment can implement the prototype	Confirm platform compatibility early.
Assumption	Indicators have sufficiently clear sources and definitions	Exclude or qualify ambiguous measures.
Dependency	Dataset access and source documentation	Required before reliable indicator selection.
Dependency	Cleaning, modelling and relationships	Required before calculations and visuals can be tested.
Dependency	Core visualisations completed	Required before meaningful usability walkthroughs.
Dependency	Instructor / reviewer availability	Affects timing and breadth of external validation.

8. Requirements Analysis and Prioritisation

Apply MoSCoW after evaluating business value, technical feasibility, data availability, dependencies, and project time. A high-value requirement is not automatically feasible: credible source coverage and compatible definitions are gating conditions.

ID	Requirement Summary	MoSCoW	Business Value	Feasibility	Dependencies / Notes
FR-01–05	Credible sources, preparation, missing-data handling, metadata and consistent definitions	Must	High	Medium; data-dependent	Source inventory and data profiling
FR-06	General City Overview	Must	High	High once data exists	Depends on minimum viable indicator set
FR-07	Five-domain organisation	Must*	High	Variable by domain	Only adequately supported domains receive populated content
FR-08, FR-10	Appropriate visuals/comparisons and clear navigation	Must	High	High	Comparison requires compatible measures
FR-11–13	Metadata, responsible interpretation and core QA	Must	High	Medium–High	Metadata availability; test cases and reviewer time
FR-09	Interactive filters	Should	Medium–High	High	Requires suitable fields and correct model relationships
UC-02 / detailed source notes	Detailed source inspection and caveats	Should	High	Medium	Source metadata must be available
Enhancement	Additional drill-through and enhanced tooltips	Could	Medium	Medium	After core pages and QA
Enhancement	Limited scenario exploration	Could	Low–Medium	Low–Medium	Only if defensible assumptions/data and validation exist
Live integration	Live government-system integration / real-time monitoring	Won't	Outside current need	Not feasible in current scope	No confirmed access or suitable live feeds
Policy automation	Automatic execution of official policy decisions	Won't	Outside prototype purpose	Out of scope	Dashboard supports human interpretation only
Advanced analytics	Predictive modelling without validated data/methods	Won't	Not required for baseline	Not justified	Requires separate scope, data and validation

*The five domains are the proposed information architecture, not a promise that all five can be populated. If a domain lacks credible, sufficiently complete data, mark it “not adequately supported by available data,” explain why, and seek instructor agreement on a reduced scope rather than inventing or substituting indicators.

Prioritisation decision rules

- Must: essential to a credible, interpretable minimum viable prototype or to basic quality and traceability.
- Should: materially improves user value but can be deferred without invalidating the core objective.
- Could: optional enhancement only after Must and Should work is stable.
- Won’t in this version: explicitly excluded capabilities; reconsider only through change control.

The project team should maintain a baseline requirements register. Reassess priorities when source review reveals missing data, incompatible geographic levels, or reporting-period gaps. Discuss significant scope reductions or additions with the instructor where appropriate.

9. Requirements Verification, Validation, and Change Control

9.1 Verification

Check that each requirement is clear, singular where possible, consistent with scope, feasible, uniquely identified, traceable to an objective or user need, and measurable through an acceptance criterion. Review terminology, duplicates, dependencies, and whether the stated verification method can produce observable evidence.

9.2 Validation

Validate that the documented requirements represent the intended academic project and anticipated user needs. Planned methods: instructor/reviewer walkthroughs; review of user stories and acceptance criteria; wireframe or prototype demonstrations; task-based testing against business needs; and a feedback log recording issue, source, decision, owner and resolution. No stakeholder sign-off is assumed.

Sign-off record — complete only if a review takes place:

Reviewer	Role	Date	Decision	Comments	Approval Status
[To complete]	[To complete]	[DD/MM/YYYY]	[Accept / Revise / Reject]	[Record comments]	[Pending / Approved / Not approved]

9.3 Requirements Traceability Matrix

Requirement ID	Requirement Summary	Related Dashboard Feature	Acceptance Test	Status
FR-01–04	Source, preparation, missing values and metadata	Source inventory / prepared model	Audit source, metadata and null treatment	Planned
FR-06–08	Overview, domains and visuals	Overview and supported domain pages	Check page coverage and valid comparisons	Planned
FR-09–10	Filters and navigation	Slicers, interactions and page navigation	Test each filter, reset and navigation path	Planned
FR-11–12	Context and responsible interpretation	Titles, tooltips, source notes and caveats	Inspect key indicators for context and overclaims	Planned
FR-13 / NFR-05	Calculation and interaction quality	Measures, relationships and report controls	Run sample reconciliation and interaction tests	Planned
NFR-03 / NFR-10	Performance and compatibility	Power BI report in target environment	Record load/interaction timings and smoke test	Planned

Status vocabulary: Planned = not yet tested; In Progress = being implemented or tested; Verified = acceptance test passed with evidence; Deferred = intentionally postponed. Update statuses only when evidence exists.

9.4 Change-Control Process

- Record the proposed change, requester, date, reason and affected requirement(s).
- Assess impact on objectives, scope, data availability, workload, schedule, quality and dependencies.
- Assign a priority and assess technical feasibility; identify trade-offs and alternatives.
- Approve or reject through the project decision process; discuss significant scope changes with the instructor where appropriate.
- Update the requirements register, IDs if needed, traceability matrix, priorities and relevant documentation.
- Implement and test affected functionality; record evidence, outcome and any remaining limitation.

A change is not considered accepted merely because it has been suggested. Keep a simple change log with ID, description, rationale, impact, decision, decision-maker, date and implementation/test status.

Final Quality-Control Checklist

Check	Review before submission
☐	Document is within the 10-page limit; cover placeholders are completed.
☐	Stakeholders, influence, interest and engagement are documented as proposed where applicable.
☐	Elicitation techniques, target sources/participants, outputs and timing are stated.
☐	Stories and use cases have testable criteria and realistic flows.
☐	Functional and non-functional requirements are specific, measurable and traceable.
☐	Constraints, assumptions, dependencies and MoSCoW priorities are consistent.
☐	Verification, validation, traceability and change control are addressed.
☐	No unperformed interviews, approvals, data coverage or tests are represented as completed.